MIL-M-28787/130A(EC)
3 May 1979
SUPERSEDING
MIL-M-28787/130(NAVY)
13 June 1974

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

NETWORK, MISCELLANEOUS, POWER, KEY CODE QPL

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic network module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Name	Network, miscellaneous
Family	Power
Key code	QPL
Size	1B
Weight	113.5 grams maximum
Failure rate	1.76 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	1.25 W

1.3 Module function. This module contains three polarized 240 µF capacitors, three polarized 180 µF capacitors, and two 6.8 kΩ resistors. The module may be used in circuits for filtering or direct current blocking where large capacitor values are required. The resistors may be used for discharging the capacitors. The capacitors are not to be used in an application where a positive voltage may occur on the negative side of the capacitor.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.76 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be QPL.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 113.5 grams.

3.5.4 Contact pins. The module shall have 20 (40 optional) contact pins positioned A1 through A20 (A21 through A40 optional).

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/130-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 Capacitance discharge. All modules shall have their capacitors discharged to 0.1 V maximum before shipping.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested shall be capable of conforming to the 25°C end-of-life limits of table I.

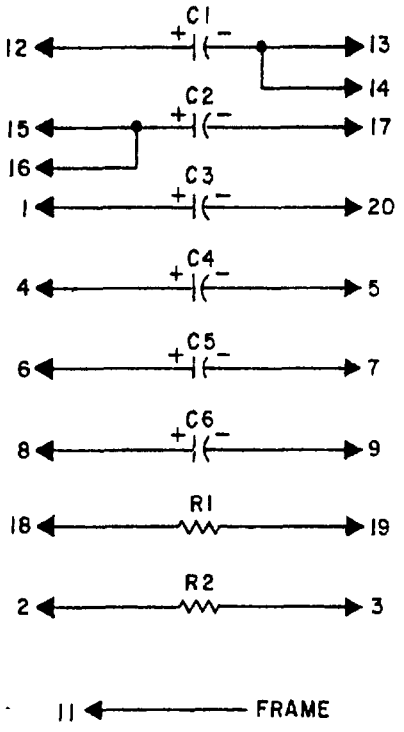
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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-45	Insulation resistance (see note 1)	II	X	1000	-	500	-	500	-	200	-	MΩ
A1A-5A	Insulation resistance (see note 3)	IIA	X	100	-	50	-	50	-	20	-	MΩ
B1-B3	DC leakage	III	X	0	30	0	30	0	75	0	75	µA
B4-B6	DC leakage	III	X	0	22	0	22	0	55	0	55	µA
C1-C5	Capacitance	IV	X	192	288	168	312	187	293	166	317	µF
C6-C8	Capacitance	IV	X	144	216	126	234	140	220	124	238	µF
D1-D5	Dissipation factor	V	S	0	0.12	0	0.18	0	0.14	0	0.21	--
D6-D8	Dissipation factor	V	S	0	0.11	0	0.165	0	0.13	0	0.195	--
E1,E2	DC resistance	VI	X	6460	7140	6120	7480	6256	7344	5780	7820	Ω
F	Continuity	VII	S	0	1	0	1	0	1	0	1	Ω

NOTES:
1. All (X) parameters shall be 100 percent tested during module acceptance tests.
2. All (S) parameters shall be 25°C ambient sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater).
3. The alternate test per table IIA (alternate) may be used in place of test per table II with the indicated limits. Either test may be run as a 100 percent test. Testing to both tests is not required.

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NOTES:
1. C1-C3 = 240 µF ±20%, 60 Vdc.
2. C4-C6 = 180 µF ±20%, 60 Vdc.
3. R1,R2 = 6.8 kΩ ±5% , 1/2 W.
4. Pins 10, 21 through 40 are not used.

FIGURE 1. QPL module function diagram, miscellaneous network.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
AC voltmeter	Item W03	1
DC power supply	Item P01	2
Power amplifier	Item A01	1
Sine wave generator	Item G01	1
DC voltmeter	Item V01	1
DC power supply	Item P02	1
DC ammeter	Item X01	1
Megohm bridge	Item R06	1
Capacitance bridge	Item C01	1
Resistance bridge	Item R02	1
Resistors		
15Ω ±10%, 4 W, variable		6
0.1Ω ±1%, 5 W		6
205Ω ±5%, 30 W		1
1.0kΩ ±10%, 5W		1
Capacitors		
≥ 10,000 µF, 75 Vdc		1

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program (SEMP) Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through F shall be performed in any order.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through F in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A through E shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

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4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. Following the vibration tests, the electrical tests of table I which are marked with an (X) in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2 for 2,000 hours. Following the life test and while still at the maximum class temperature, electrical tests B through E shall be performed in the order specified in 4.4.2 using the maximum class temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through F shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life.

6.3 Cross reference of classification. Part number M28787/130-1 supersedes part number M28787/130. The two parts are interchangeable.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N027)

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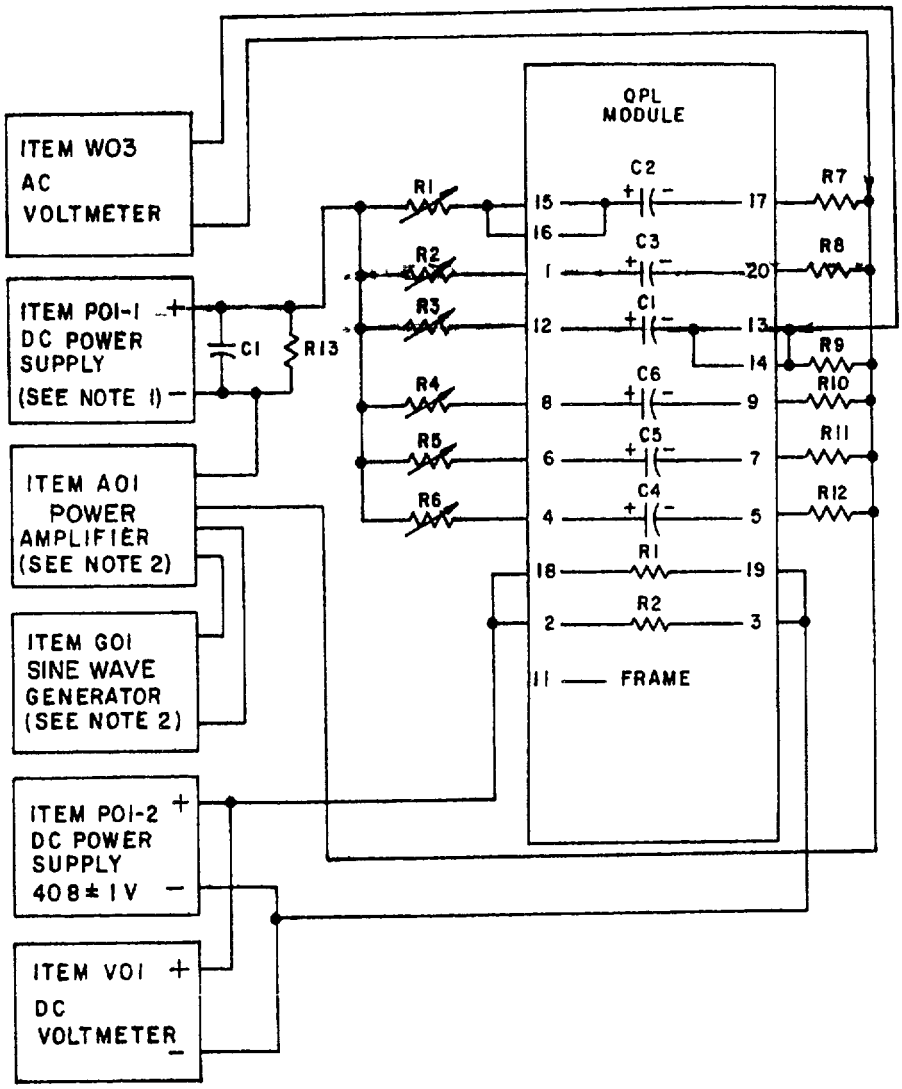


FIGURE 2. Operating temperature, component temperature, and life test circuit.

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NOTES: 1. The ripple voltage shall be superimposed upon a direct current voltage equal to 2/3 of the specified working voltage limit.

2. An alternate power source such as a variac or powerstat and stepdown transformer with 60 Hz line input power may be used. The output voltage shall be adjusted so that the requirements of note 4 can be met.

3. Capacitors shall be energized with the following.

   (a) Ripple current rating at 60 Hz shall be 440 mArms for C2 (pins 15 and 16, to 17), C3 (pin 1 to 20), and C1 (pin 12 to 13, and 14)

   (b) Ripple current rating at 60 Hz shall be 330 mArms for C6 (pin 8 to 9), C5 (pin 6 to 7), and C4 (pin 4 to 5).

4 Adjust R1 through R6 per notes 3 (a) and 3 (b), using a voltage drop across resistors R7 through R12 to determine the current.

5. R1-R6 = 15Ω ±10%, 4 W, variable.

6 R7-R12 = 0.1Ω ±1%, 5 W.

7. R13 = 205Ω ±5%, 30 W

8. C1 ≥ 10,000 µF, 75 Vdc

FIGURE 2. Operating temperature, component temperature, and life test circuit - Continued.

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TABLE II. Insulation resistance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Insulation resistance R2-C3	2,3	1,20	1,000	∞	MΩ	3
A2	Insulation resistance C4-C3	4,5	1,20				
A3	Insulation resistance C4-R2	4,5	2,3				
A4	Insulation resistance C5-C3	6,7	1,20				
A5	Insulation resistance C5-R2	6,7	2,3				
A6	Insulation resistance C5-C4	6,7	4,5				
A7	Insulation resistance C6-C3	8,9	1,20				
A8	Insulation resistance C6-R2	8,9	2,3				
A9	Insulation resistance C6-C4	8,9	4,5				
A10	Insulation resistance C6-C5	8,9	6,7				
A11	Insulation resistance C1-C3	12,13, 14	1,20				
A12	Insulation resistance C1-R2	12,13, 14	2,3	1,000	∞	MΩ	3

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TABLE II. Insulation resistance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A13	Insulation resistance C1-C4	12,13, 14	4,5	1,000	∞	MΩ	3
A14	Insulation resistance C1-C5	12,13, 14	6,7				
A15	Insulation resistance C1-C6	12,13, 14	8,9				
A16	Insulation resistance C2-C3	15,16, 17	1,20				
A17	Insulation resistance C2-R2		2,3				
A18	Insulation resistance C2-C4		4,5				
A19	Insulation resistance C2-C5		6,7				
A20	Insulation resistance C2-C6		8,9				
A21	Insulation resistance C2-C1	15,16, 17	12,13, 14				
A22	Insulation resistance R1-C3	18,19	1,20				
A23	Insulation resistanc R1-R2	18,19	2,3				
A24	Insulation resistance R1-C4	18,19	4,5	1,000	∞	MΩ	3

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TABLE II. Insulation resistance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A25	Insulation resistance R1-C5	18,19	6,7	1,000	∞	MΩ	3
A26	Insulation resistance R1-C6		8,9				
A27	Insulation resistance R1-C1		12,13, 14				
A28	Insulation resistance R1-C2	18,19	15,16, 17				
A29	Insulation resistance Frame C3	11	1,20				
A30	Insulation resistance Frame R2		2,3				
A31	Insulation resistance Frame-C4		4,5				
A32	Insulation resistance Frame-C5		6,7				
A33	Insulation resistance Frame-C6		8,9				
A34	Insulation resistance Frame-C1		12,13, 14				
A35	Insulation resistance Frame-C2		15,16, 17				
A36	Insulation resistance Frame-R1	11	18,19	1,000	∞	MΩ	3

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TABLE II. Insulation resistance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A37	Insulation resistance Frame-*unused pins	11	10, 21-40	1,000	∞	MΩ	3
A38	Insulation resistance *Unused pins-C3	10, 21-40	1,20				
A39	Insulation resistance *Unused pins-R2		2,3				
A40	Insulation resistance *Unused pins-C4		4,5				
A41	Insulation resistance *Unused pins-C5		6,7				
A42	Insulation resistance *Unused pins-C6		8,9				
A43	Insulation resistance *Unused pins-C1		12,13, 14				
A44	Insulation resistance *Unused pins-C2		15,16, 17				
A45	Insulation resistance *Unused pins-R1	10, 21-40	18,19	1,000	∞	MΩ	3

* Pins 21 through 40 are optional; if present, connect as shown.

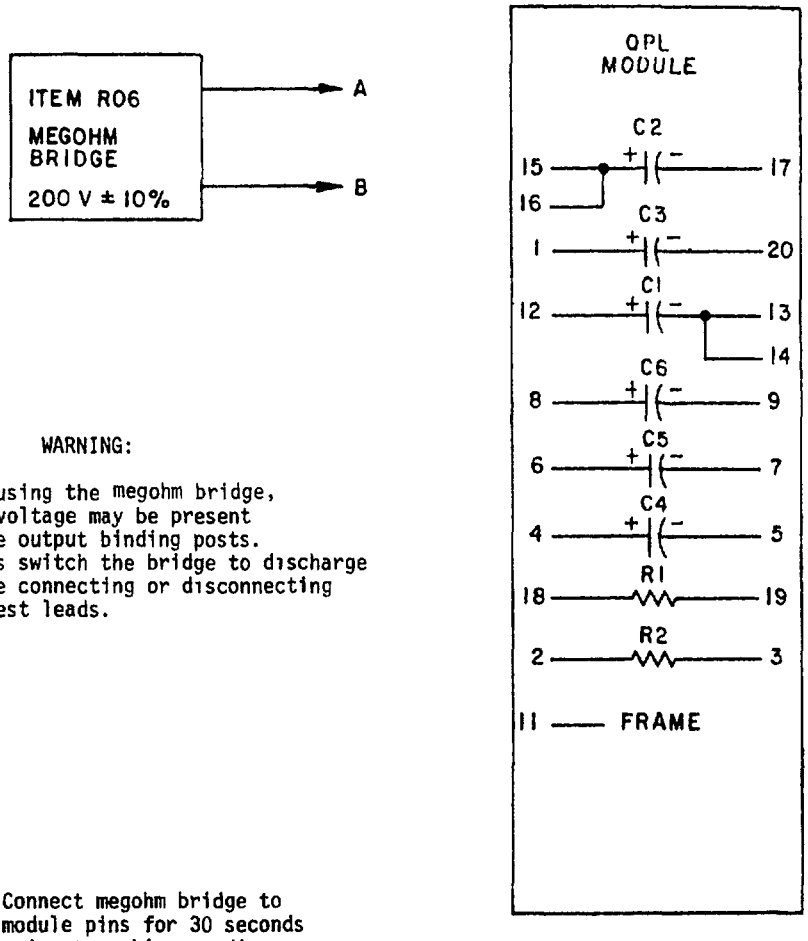
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TABLE IIA. Alternate insulation resistance test.

Test condition	Parameter	Connect		Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1A	Insulation resistance	2-9	1, 12-20	100	∞	MΩ	3
A2A	Insulation resistance	2-5, 12-14, 18,19	15-17, 20				
A3A	Insulation resistance	1-3, 6,7, 18-20	4-5, 8,9, 12-17				
A4A	Insulation resistance (see note)	1-9, 11-20	10, 21-40				
A5A	Insulation resistance	11	1-9, 12-20	100	∞	MΩ	3

NOTE: Pins 21 through 40 are optional; if present, connect as shown.

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WARNING:

When using the megohm bridge, high voltage may be present on the output binding posts. Always switch the bridge to discharge before connecting or disconnecting the test leads.

NOTE Connect megohm bridge to module pins for 30 seconds prior to taking readings. In addition, there shall be no momentary or intermittent arcing or other indication of breakdown or visible evidence of damage.

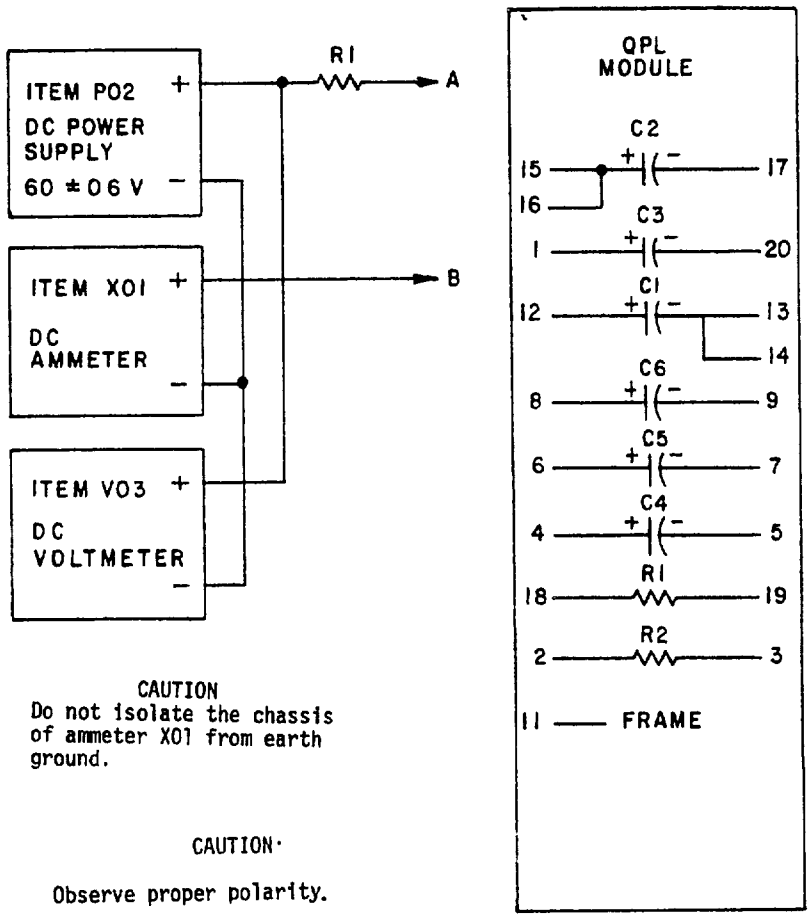
FIGURE 3. Insulation resistance test circuit for table II and table IIA (alternate).

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TABLE III. DC leakage test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	DC leakage (C2)	15	17	0	30	µA	4
B2	DC leakage (C3)	1	20		30		
B3	DC leakage (C1)	12	13		30		
B4	DC leakage (C6)	8	9		22		
B5	DC leakage (C5)	6	7		22		
B6	DC leakage (C4)	4	5	0	22	µA	4

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NOTES:
1. R1 = 1 kΩ ±10%, 5 W.
2. Unit conforms to requirements if leakage drops below limit indicated in table III in five minutes or less.

FIGURE 4. DC leakage test circuit for table III.

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TABLE IV. Capacitance test.

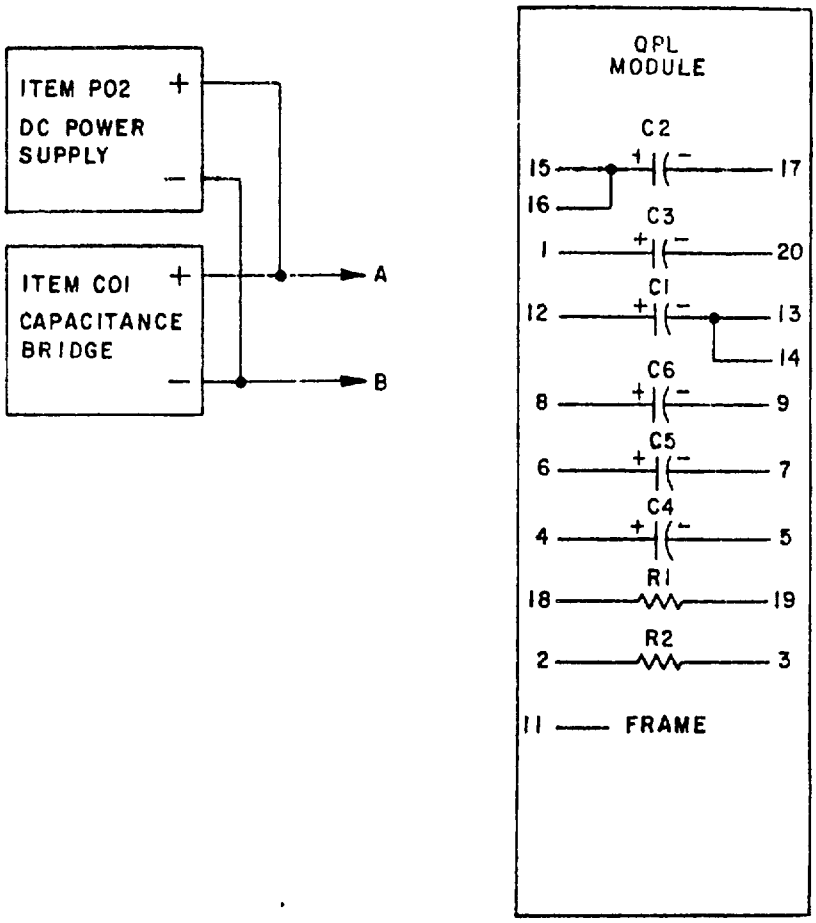
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Capacitance (C2)	15	17	192	288	µF	5
C2	Capacitance (C2)	16	17	192	288		
C3	Capacitance (C3)	1	20	192	288		
C4	Capacitance (C1)	12	13	192	288		
C5	Capacitance (C1)	12	14	192	288		
C6	Capacitance (C6)	8	9	144	216		
C7	Capacitance (C5)	6	7	144	216		
C8	Capacitance (C4)	4	5	144	216	µF	5

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TABLE V. Dissipation factor test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Dissipation factor (C2)	15	17	0	0.12	--	5
D2	Dissipation factor (C2)	16	17				
D3	Dissipation factor (C3)	1	20				
D4	Dissipation factor (C1)	12	13				
D5	Dissipation factor (C1)	12	14		0.12		
D6	Dissipation factor (C6)	8	9		0.11		
D7	Dissipation factor (C5)	6	7		0.11		
D8	Dissipation factor (C4)	4	5	0	0.11	--	5

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NOTES:
1. Circuit shown for GR capacitance bridge.
2. The ac voltage shall not exceed 0.707 Vrms at 120 ±6 Hz; this shall be superimposed on a polarizing voltage of 1.4 ±0.1 Vdc.
3. For test conditions D1-D8, the total test circuit load resistance shall not exceed 50 milliohms.

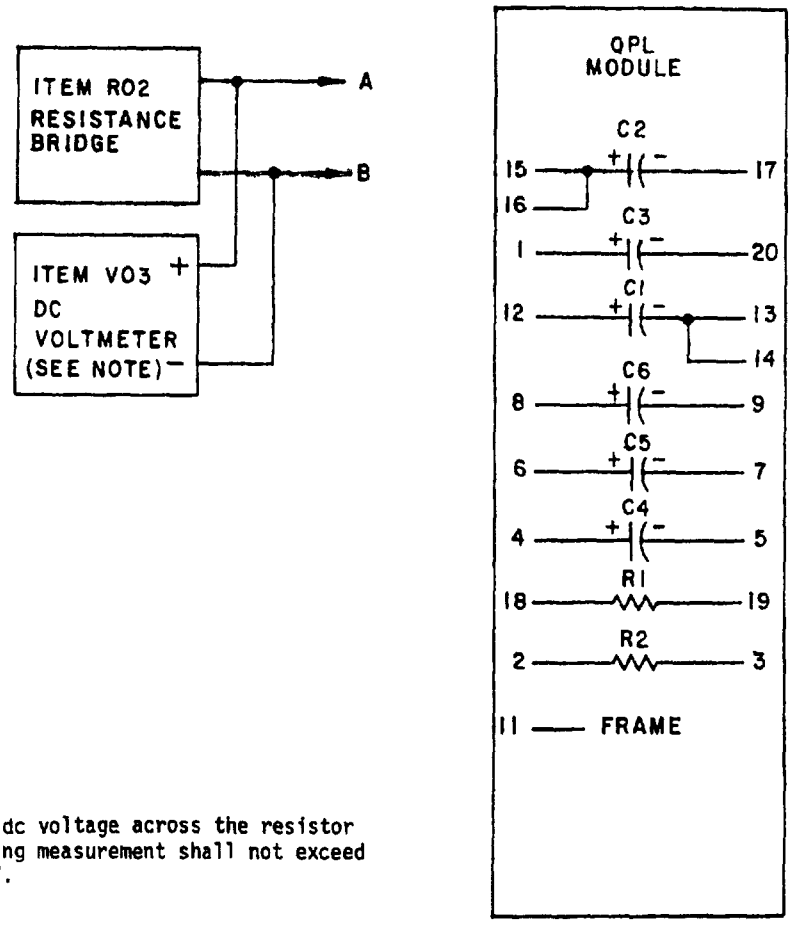
FIGURE 5. Capacitance and dissipation factor test circuit for tables IV and V.

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TABLE VI. DC resistance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	DC resistance (R1)	18	19	6.46	7.14	kΩ	6
E2	DC resistance (R2)	2	3	6.46	7.14	kΩ	6

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NOTE: The dc voltage across the resistor during measurement shall not exceed 10 V.

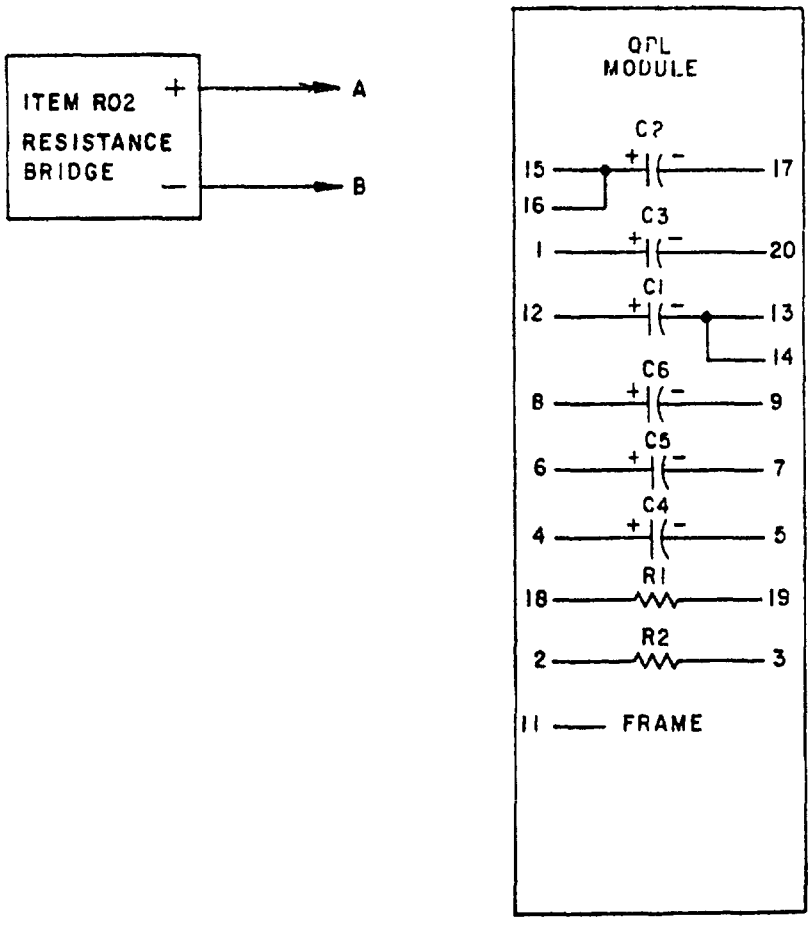
FIGURE 6. DC resistance test circuit for table VI.

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TABLE VII. Continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F	Continuity	11	Frame	0	1	Ω	7

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NOTE: Penetrate the anodized finish with a sharp pointed probe to obtain electrical contact.

FIGURE 7. Continuity test circuit for table VII

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

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